

Zhiyi Chen

Tel: +1 (470) 659-3346 Email: zchen798@gatech.edu
Klaus Advanced Computing Building, Room 3337, Atlanta, GA, USA

Education

Georgia Institute of Technology

Ph.D. Candidate in Computer Science; Advisor: Alberto Dainotti

Master of Science in Computer Science; Concentration: Machine Learning

GPA: 4.0/4.0

Atlanta, GA

Jan 2021 - present

Jan 2021 - May 2025

Tsinghua University

Bachelor of Science in Computer Science and Technology

GPA: 3.73/4.0

Beijing, China

Aug 2016 - Jul 2020

Selected Research Experience

Internet Measurement + LLM: Leveraging Web Presence and AI to Improve AS Classification

Georgia Tech

Research Assistant; Advisors: Alberto Dainotti, Cecilia Testart, Zachary S. Bischof

Nov 2024 - Dec 2025

- Developed a novel methodology for AS business classification by leveraging LLMs and AS web presence.
- Doubled Micro-F1 over the ASdb baseline (from 0.40 to 0.80) and improved exact-match accuracy by ~50%.

Work Experience

Google LLC

Remote

Student Researcher; Advisors: Alex Krentsel, Sylvia Ratnasamy

Jan 2026 - present

- Building an AI agent system for analyzing outages in network infrastructure.

ByteDance, Inc.

San Jose, CA

Research Intern; Advisors: Peng Li, Tieying Zhang

May 2025 - Aug 2025

- Designed an automated prompt-engineering AI agent system that improves LLM performance on text-to-SQL tasks in complex, real-world settings.
- Reduced average syntax-correction rounds from 3.14 to 1.65 ($\approx 47\%$) and raised the semantic correctness pass rate from 18.8% to 51.6% (+32.8 percentage points, $\sim 2.7\times$) on the Spider 2.0 (Snowflake) benchmark.

Catchpoint Systems, Inc.

Remote

CTG Intern; Advisor: Gavin Lynch

Jun 2024 - Aug 2024

- Explored complex Internet test data from test nodes all over the world towards cloud and content providers.
- Designed automatic approaches for detecting false positives in inferred cloud outages and developed methods for detecting cloud performance degradations.

Coalition, Inc.

Remote

Research Intern; Advisors: Daniel Woods, Morgan Hervé-Mignucci

Jun 2023 - Aug 2023

- Leveraged public Internet measurement platforms and GIS tools to infer and visualize geographic locations of data centers that host policyholder domains.
- Explored the aggregation of cyber risks on physical infrastructure, especially critical points such as datacenters.

Publications

- **Chen, Z.**, Bischof, Z. S., Testart, C., Dainotti, A. "AS2Biz: Leveraging Web Presence and AI to Improve AS Business Classification", accepted by Internet Measurement Conference (IMC) 2026.
- **Chen, Z.**, Bischof, Z. S., Testart, C., Dainotti, A. **Improving the Inference of Sibling Autonomous Systems**, published in the Passive and Active Measurement Conference (PAM) 2023.
 - Our dataset won *PAM 2023 Best Community Artifact Award*.
 - A lightning talk presented at North American Network Operators' Group (NANOG) 87 in Atlanta.
- Li, P., **Chen, Z.**, Chu, X., Rong, K. **DiffPrep: Differentiable Data Preprocessing Pipeline Search for Learning over Tabular Data**, published in The ACM Special Interest Group on Management of Data (SIGMOD) 2023.
- **Chen, Z.**, Lin, T. **Principal Gradient Direction and Confidence Reservoir Sampling for Continual Learning**, published in the International Conference on Artificial Neural Networks (ICANN) 2021.